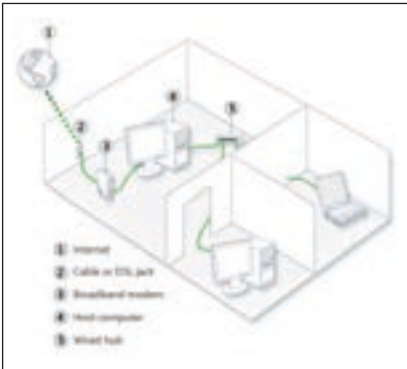


of them have different capabilities and serve different purposes. Let's explore each.

Hubs

In their bare essence, hubs enable computers on a network to communicate. An ethernet cable is used to plug each computer into the hub, and



Network with a Hub

information is then sent from one computer to another through the hub. One of the glaring drawbacks of a hub is its inability to identify the source or intended destination of the information it receives. As a consequence, a hub ends up sending the information it receives to all of the computers connected to it, including the one that sent it. Moreover, a hub can send or receive information, but it

can't do both at the same time. This makes hubs slower than switches. On the upside, hubs are the least complex and the least expensive of these devices.

Technically speaking, three different types of hubs exist:

- ▶ **Passive hubs:** Do not amplify the electrical signal of incoming packets before broadcasting them out to the network.
- ▶ **Active hubs:** Perform this amplification. Another dedicated network device which is capable of amplifying the electric signals is a repeater. Some people use the terms “concentrator” when referring to a passive hub and “multiport repeater” when referring to an active hub.
- ▶ **Intelligent hubs:** Used to add extra features that are of particular importance to businesses to an active hub. An intelligent hub typically is stackable, i.e. it's built in such a way that multiple units can be placed one on top of another to conserve space. It also typically includes remote management capabilities via SNMP and virtual LAN (VLAN) support.

Switches

The working principle behind a switch is somewhat similar to that of a hub's. However, switches are capable of identifying the intended destination of the